

Antibacterial activity of a substance produced by the fungus *Pycnoporus sanguineus* (Fr.) Murr.

A. Smânia^{*a}, F. Delle Monache^b, E.F.A. Smânia^a, M.L. Gil^a, L.C. Benchetrit^c,
F.S. Cruz^c

^aDepartamento de Microbiologia e Parasitologia, Universidade Federal de Santa Catarina, 88040-900 — Florianópolis SC, Brazil

^bIstituto di Chimica, Università Cattolica, Rome, Italy

^cInstituto de Microbiologia, Universidade Federal, Rio de Janeiro, Brazil

Received 22 September 1994; revision received 15 November 1994; accepted 17 November 1994

Abstract

A fraction obtained from the culture fluids of *Pycnoporus sanguineus* fungus was shown to contain a compound with biological activity against strains of *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Salmonella typhi*, *Staphylococcus aureus* and members of the genus *Streptococcus*. The fraction was clearly more active on Gram-positive cocci than on Gram-negative bacilli.

Keywords: *Pycnoporus sanguineus*; Antibacterial activity; Phenoxazinones pigments

1. Introduction

Many pharmaceutical products are of plant, bacterial or fungal origin, although synthetic organic compounds are substantially used and have proved their value in a broad range of human infections. Since the introduction of penicillin (from *Penicillium notatum*; Fleming, 1929) into therapeutics more than 50 years ago, numerous drugs with antibacterial activity have been obtained from the Actinomycetes and *Bacillus*.

Pycnoporus sanguineus is a slow-growing saprophytic fungus belonging to the Basidiomycetes of the family *Polyporaceae*, which causes decay of

certain types of wood in the forests of tropical and subtropical areas of the northern and southern hemispheres (Nobles and Frew, 1962). This fungus has long been used in popular medicine by indigenous tribes of the Americas and Africa for treatment of a number illnesses (Alibert, 1944; Fidalgo, 1965; Fidalgo and Hirata, 1979; Pérez-Silva et al., 1988) and also by certain rural communities of the State of Santa Catarina, south of Brazil, for treatment of skin lesions.

This paper explains investigations into the ability of a substance extracted from *Pycnoporus* culture fluids to inhibit the growth of various bacterial species. In addition to dealing with an aspect of fungus biochemistry worthy of investigation, this study also falls well within the topic of

^{*} Corresponding author.

biological diversity defined by the United Nations Conference on Environment and Development (UNCED) in Rio de Janeiro, as a challenging problem requiring positive action within the next few years, especially in developing countries (UNCED, 1992).

2. Materials and methods

2.1. Fungal strain and culture conditions

The wild-growing fungus strain (no. MIP 89007) isolated originally from a decaying forest tree trunk in the southern State of Santa Catarina, Brazil, was grown on a plate containing potato-dextrose agar. After which, this cultured fungus (about 1 cm² of mycelium) was inoculated in 400 ml of broth, and growth of the mycelial phase was allowed to proceed at 25°C for 10 days. Amounts (15 ml) of the culture-broth were transferred to 20 flasks containing 150 ml of the same medium and the cultures were incubated at 25°C for 20 days. Thereafter, the culture fluids were lyophilized to yield 72.15 g of a dark brownish powder (starting material). Broth for fungal growth was an infusion of peeled potatoes (200 g/liter) at pH 6.0 with 2% glucose. Agar for maintenance of stock cultures of the fungus contained the broth and 1.5% agar.

2.2. Extraction and chromatographic systems

The lyophilized fungal material was placed in a Soxhlet apparatus and exhaustively extracted with *n*-hexane and then acetone to yield 383 mg and 938 mg extracts, respectively. A part of the acetone residue (818 mg) was chromatographed on a silica gel column using the following eluents of increasing polarity: ethyl acetate/methanol/acetic acid (98.5:1:0.5–88:8:4) and methanol. Fractions of 10 ml were collected, and the composition of each was examined on silica gel TLC plates (60f₂₅₄, Merck, 2 mm thick) with ethyl acetate/acetic acid (9:1) as the developer. Chromatograms were visualized under UV light (254 and 366 nm) followed by spraying with sulfuric acid and heating. Authentic samples of phenoxazin-3-one type pigments also were spotted together on TLC plates. Fractions which showed similar patterns were pooled and tested for antibacterial activity.

2.3. Bacterial strains and detection of antimicrobial activity

The study was carried out against the following microorganisms: *Escherichia coli* ATCC 25922 (American Type Culture Collection, Rockville, MD), *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* ATCC 27853, *Salmonella typhi*, *Staphylococcus aureus* ATCC 25923, and Groups A, B, C and G Streptococci. The indicator strains were incubated at 36°C for 18 h. The cultures were then diluted into broth to contain 10⁶ CFU/ml. Mueller-Hinton agar and broth (Difco Laboratories) were used for bacterial growth (5% defibrinated horse blood was added when required). All of the bacterial cultures were incubated under aerobic conditions.

2.4. Agar diffusion method

Cotton swabs charged with each bacterial suspension (10⁶ CFU/ml) were inoculated on plates and bacteria were spread evenly over the surface of the agar medium. Wells of 7 mm diameter were aseptically cut and filled with different amounts of the extracts and chromatographic fractions. The plates were incubated at 36°C for 18 h and examined for inhibition of the growth of the indicator strains. A positive result was defined as a zone of 9 mm or more in diameter of inhibited growth of the bacterial strain and was interpreted as confirming production of an antibacterial substance by the fungus.

2.5. Estimation of minimal inhibitory concentrations (MIC)

The estimations were carried out in liquid media which contained decreasing amounts of the most active fraction (Jones et al., 1985). The MIC was defined as the lowest concentration of a fraction which was able to completely inhibit the growth of each bacterial strain. Serial two-fold dilutions of the fraction were prepared in broth. To each test tube was added an equal volume of the adjusted inoculum. After incubation at 36°C for 18 h the MIC was read. A subculture on agar plates was made from each of the tubes without apparent growth to determine the minimal bactericidal concentration (MBC; Schoenknecht et al., 1985; Men-

sah et al. 1990). The MBC was the lowest concentration of the fraction which yielded a >99.9% reduction in the number of colony forming units (CFU).

3. Results

3.1. Antibacterial activity

Table 1 shows the results obtained by diffusion tests during the extraction and purification stages. The starting material only inhibited the growth of the strain of *Staph. aureus* when 8 mg was used. Fraction B (FB) showed the best antibacterial activity. It was approximately 88 times more active than the starting material and 2.8 times more active than the acetone extract. The hexane extract and Fraction D (FD) did not inhibit the growth of the bacterial strains when 12 mg and 2.5 mg were used, respectively.

The in vitro broth dilution method showed that FB was inhibitory against all the microorganisms tested (Table 2). The lowest MIC values were obtained against the Streptococci strains. The highest values were obtained against Gram-negative bacteria. In general, for FB, MBCs approximated or were close to the MICs. The MBCs were equal to the MICs for *E. coli*, *Streptococcus pyogenes* (Group A Streptococci), *Pseudomonas aeruginosa*, *Salmonella typhi* and *Staph. aureus*. The MBC/MIC ratio was 2 for the other bacteria tested.

Table 2

MIC and MBC values of FB for various microorganisms

Microorganisms	MIC	MBC
<i>E. coli</i>	0.625 ^a	0.625
<i>K. pneumoniae</i>	0.625	1.250
<i>P. aeruginosa</i>	1.250	1.250
<i>Salmonella typhi</i>	0.312	0.312
<i>Staph. aureus</i>	0.156	0.156
Group A Streptococci	0.039	0.039
Group B Streptococci	0.039	0.078
Group C Streptococci	0.019	0.039
Group G Streptococci	0.039	0.078

^aExpressed in mg/ml of Fraction B.

3.2. Fractionation of the acetone extract

Chromatography of the acetone extract by silica gel column guided by R_f in TLC (ethyl acetate/acetic acid, 9:1) allowed for the separation of four fractions, designated as FA, FB, FC and FD. These fractions corresponded to 13.93%, 23.11%, 5.01% and 46.57% of the material obtained by fractionation, respectively. FB which possessed the highest antimicrobial activity was characterized chemically.

3.3. Chemical characterization of FB

Examination of FB by analytical TLC resulted in the visualization of a strongly-colored band and

Table 1

Antibacterial activity of the acetone extract and fractions obtained by silica gel column against *Staph. aureus*

Test material	Zone of inhibition (mm)								
	2.5 ^a	1.5	1.0	0.75	0.5	0.25	0.12	0.09	0.06
Hexane extract	NA ^b	NA	NA	NA	NA	NA	NA	NA	NA
Acetone extract	22 ^c	21	19	17	17	10	NA	NA	NA
Fraction A	17	16	11	10	NA	NA	NA	NA	NA
Fraction B	27	25	25	22	20	15	13	9	NA
Fraction C	16	NA	NA	NA	NA	NA	NA	NA	NA
Fraction D	NA	NA	NA	NA	NA	NA	NA	NA	NA

^aAmount (mg) contained in each well.

^bNA, complete lack of activity.

^cZones of inhibition measured at 18 h.

Table 3
 R_f values of FB major constituent in various solvent systems

Systems ^a	R_f values
A	0.78
B	0.54 ^b
C	0.48 ^b
D	0.10
E	0.78

^aSystem A: ethyl acetate/acetic acid (9:1);

System B: benzene/ethyl acetate/formic acid (6:3:1);

System C: benzene/ethyl acetate/acetic acid (6:3:1);

System D: chloroform/acetic acid (97:3);

System E: ethyl acetate/methanol/acetic acid (95:4:1).

^bSullivan and Henry (1971).

two other weakly colored bands. The R_f values of the major constituent in different chromatographic systems are listed in Table 3.

Ultraviolet spectrum (Lambda 5 UV/VIS spectrophotometer) of FB in dioxane exhibited: λ_{max} (log ϵ) = 264 (4.6) nm, 430 (4.4) nm, 452 (sh) (4.3) nm.

4. Discussion

The results obtained in this work show that among the fractions obtained from *P. sanguineus* by chromatography using a silica gel column, one is especially active (Table 1). FB exerted an inhibitory effect in vitro on all microorganisms tested. In general, however, there was higher inhibitory activity for Gram-positive than Gram-negative bacteria. The MICs and MBCs ranging for the first group were 0.019–0.156 and 0.039–0.156, and for the second group was 0.625–1.250, for the two tests. Members of the genus *Streptococcus* appeared to be the most sensitive organisms. The ranges of MICs and MBCs were 0.019–0.039 and 0.039–0.078, respectively. The MBCs and MICs ratio for the nine strains tested were either the same (i.e., ratio = 1) or differed by one dilution (ratio = 2) (Table 2).

Antimicrobial activity with related fungi has been already studied. Bose (1946) extracted *Polystictus sanguineus* with various solvents and observed that those extracts obtained by using

ether, chloroform and ethyl acetate were active against *Salmonella* spp., but no activity was found for acetone extracts. In contrast, Lemberg (1952) found no evidence for antibiotic action against Gram-negative organisms when he tested ether and chloroform extracts. In the present investigation, extracts obtained with less polar solvents (i.e., butyl alcohol, dichloromethane, ethyl ether, petroleum ether, heptane and hexane) did not inhibit the growth of the bacterial strains at a concentration of 100 mg/ml, and extracts obtained with more polar solvents (i.e., ethyl acetate, acetone and methanol) showed an activity at lower concentrations (unpublished data).

A comparison of chemical and spectroscopic data obtained here with those published previously (Cavill et al., 1957; Gripenberg, 1963; Ayyangar et al., 1981; Achenbach and Blümm, 1991), indicates that the major constituent of FB is a phenoxazin-3-one type pigment. This constituent showed identical R_f values with those described by Sullivan and Henry (1971) in connection with samples of cinnabarin (Table 3).

In summary, it may be concluded that the results obtained in this study could partly justify the popular use of *P. sanguineus* for the treatment of various infectious diseases. For this reason, studies related to the secondary metabolites of this fungus and their biological activity are presently being undertaken.

Acknowledgments

The authors wish to thank Dr. Clarice L. Leite (Department of Botany, Federal University of Santa Catarina) for identification of the fungus, and Dr. Hans Achenbach (Institut für Pharmazie und Lebensmittelchemie, Universität Erlangen, Germany) for providing authentic samples of the phenoxazin-3-one type pigments. This study was supported by the Universidade Federal de Santa Catarina, Conselho de Ensino para Graduados (CEPG), Consiglio Nazionale delle Ricerche (CNR), Financiadora de Estudos e Projetos (FINEP), Fundação Universitária José Bonifácio (FUJB). During the studies, Dr. Artur Smânia Jr. was the recipient of grants from Coordenação de

Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq). Drs. Leslie C. Benchetrit and Fernando S. Cruz are Senior Investigators of CNPq.

References

- Achenbach, H. and Blümm, E. (1991) Investigation of the pigments of *Pycnoporus sanguineus* — Metabolites pycnosanguin and new phenoxazin-3-ones. *Archiv der Pharmazie (Weinheim)* 324, 3–6.
- Alibert, H. (1944) Note sur les champignons poussant dans le bas Dahomey et sur deux agaricinées estimées des indigènes de cette même région. *Notes Africaines* 22, 11–12.
- Ayyangar, N.R., Lugade, A.G., Sane, M.G. and Srinivasan, K.V. (1981) Separation of 2-amino-3h-phenoxazin-3-one impurity from *o*-amino-phenol. *Journal of Chromatography* 208, 113–116.
- Bose, S.R. (1946) Antibiotics in a Polyporus (*Polystictus sanguineus*). *Nature* 158, 292–296.
- Cavill, G.W.K., Clezy, P.S. and Tetaz, J.R. (1957) The chemistry of mould metabolites. Part II. A partial structure for polystictin. *Journal of the Chemical Society* 2646–2651.
- Fidalgo, O. (1965) Conhecimento micológico dos índios brasileiros. *Rickia* 2, 1–10.
- Fidalgo, O. and Hirata, J.M. (1979) Etnomicologia caiabi, txicão e txucarramãe. *Rickia* 8, 1–5.
- Fleming, A. (1929) On the antibacterial action of cultures of a *Penicillium*, with special reference to their use in isolation of *B. influenzae*. *British Journal of Experimental Pathology* 10, 226–236.
- Gripenberg, J. (1963) Fungus pigments. XIII: Tramesanguin, the pigment of *Trametes cinnabarina* var. *sanguinea* (L.) Pilat. *Acta Chemica Scandinavica* 17, 703–708.
- Jones, R.N., Barry, A.L., Gavan, T.L. and Washington II, J.A. (1985) Microdilution and macrodilution broth procedures. In: E.H. Lennette, A. Balows, W.J. Hausler, Jr. and H.J. Shadomy (Eds.), *Manual of Clinical Microbiology*. American Society for Microbiology, Washington, DC, pp. 972–977.
- Lemberg, R. (1952) Nitrogenous pigments from the fungus *Coriolus sanguineus* (*Polystictus cinnabarinus*). *Australian Journal of Experimental Biology and Medical Science* 30, 271–278.
- Mensah, J.L., Lagarde, I., Ceschin, C., Michael, G., Gleye, J. and Fourast, I. (1990) Antibacterial activity of the leaves of *Phyllanthus discoideus*. *Journal of Ethnopharmacology* 28, 129–133.
- Nobles, M.K. and Frew, B.P. (1962) Studies in wood-inhabiting Hymenomycetes. V: The genus *Pycnoporus* Karst. *Canadian Journal of Botany* 40, 987–1016.
- Pérez-Silva, E., Aguirre-Acosta, E. and Pérez-Amador, C. (1988) Aspectos sobre el uso y distribución de *Pycnoporus sanguineus* (Polyporaceae) en México. *Revista Mexicana de Micología* 4, 137–144.
- Schoenknecht, F.D., Sabath, L.D. and Thornsberry, C. (1985) Susceptibility tests: special tests. In: E.H. Lennette, A. Balows, W.J. Hausler, Jr. and H.J. Shadomy (Eds.), *Manual of Clinical Microbiology*. American Society for Microbiology, Washington, DC, pp. 1000–1008.
- Sullivan, G. and Henry, E.D. (1971) Occurrence and distribution of phenoxazinone pigments in the genus *Pycnoporus*. *Journal of Pharmaceutical Sciences* 60, 1097–1098.
- UNCED (1992). United Nations Conference on Environment and Development, A/CONF.151/26.